



- 5 (a) Show that the curve with Cartesian equation

$$(x^2 + y^2)^2 = 6xy$$

has polar equation $r^2 = 3 \sin 2\theta$.

[2]

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The curve C has polar equation $r^2 = 3 \sin 2\theta$, for $0 \leq \theta \leq \frac{1}{2}\pi$.

- (b) Sketch C and state the maximum distance of a point on C from the pole.

[3]





[2]

[illegible]

[6]

[illegible]